A LOCALLY FINITE TYPE SURJECTIVE MORPHISM OF ALGEBRAIC SPACES IS SURJECTIVE ON FINITE TYPE POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short but complete: it builds the point on the fibre and transports it back, using the base change stability of local finite typeness in between.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Finite type points of algebraic spaces and their behaviour under locally finite type morphisms. Lemma lemma-finite-type-points-morphism, used at proof lines 5115-5116 and 5126-5128. Gives one inclusion and transports the point found on X_k back to X.
- Nonempty algebraic spaces of finite type over a field have a finite type point. Lemma lemma-identify-finite-type-points, used at proof lines 5120-5122. Produces the point x of X_k.
- Local finite typeness is stable under base change. Lemma lemma-base-change-finite-type, used at proof lines 5123-5125. Makes X_k -> X locally of finite type.
- Representation of a point by a locally finite type morphism from a field. used at proof lines 5117-5118. Supplies the field-valued test point for y.

Source reference. Original lemma 03XH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5148–5148. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06EI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8217–8217. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06EH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8216–8216. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite type and surjective, then $f(X_{ft\text{-}pts}) = Y_{ft\text{-}pts}$.*

Proof. We have $f(X_{\text{ft-pts}}) \subset Y_{\text{ft-pts}}$ by Lemma 06EI. Let $y \in |Y|$ be a finite type point. Represent y by a morphism $\text{Spec}(k) \to Y$ which is locally of finite type. As f is surjective the algebraic space $X_k = \text{Spec}(k) \times_Y X$ is nonempty, therefore has a finite type point $x \in |X_k|$ by Lemma 06EH. Now $X_k \to X$ is a morphism which is locally of finite type as a base change of $\text{Spec}(k) \to Y$ (Lemma 03XH). Hence the image of x in X is a finite type point by Lemma 06EI which maps to y by construction. □